RESEARCH TO SOLUTION

SetuTrace

A student-centred proposal for trustworthy institutional reporting, traceable corrections and authorised verification.

Evidence-led design / Complete proposed technology stack / Validation gates

Prepared 27 October 2026

Built from the team's GitHub and Notion research, supplied uploads, specialist agent research and a reproduced synthetic experiment.

Dated research/design report; the repository now includes a separately tested synthetic prototype. No live government integration, operational deployment, institutional endorsement or field validation is claimed.

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Links in the report open primary publications or the main-branch research files. The source ledgers record retrieval dates, sections and limits.

A correctable evidence trail for student reporting

Idea report · SIH26263 supplied brief · Completed 27 October 2026

Prepared by a separate AI design-thinking innovator after the six research workstreams were assembled. Human workstream ownership remains as recorded in team.json; Kanishkar B is the human team lead. Research began 26 October. This is the 27 October research proposal for human review. At that stage no application had been implemented. The 28 October submission prototype now implements a focused subset; see actual test evidence. No provider credentials, live government integration, interviews or field pilot are claimed. SetuTrace is a working name, with no claim of official affiliation or name clearance.

Recommendation: build a narrow reporting and correction layer alongside institutional systems. Its useful output is an explainable submission snapshot: which enrolment records were counted, which approval and rule versions were used, what remains unresolved, and who can correct it. APAAR helps connect authorised evidence; it does not become a universal admission verdict.

The ten requirements come from the supplied statement. The official SIH page remained inaccessible during research, so final wording, event dates and submission conditions are **unverified**. Resolve this before submission; do not substitute a third-party listing for organiser confirmation. Official verification record.

1. The problem worth solving

The problem is not simply that colleges lack software. Samarth already documents student imports, validation, allocation and edits. NAD already supports issuer-published academic awards and authorised sharing. AISHE collects programme-level intake and enrolment aggregates. These sources establish existing capabilities; they do not establish how consistently a particular institution reconciles them. Samarth STU-001, GOP-103; NAD FAQ, SUD-103; workflow research, GOP-110.

The proposed gap is the accountable handoff between a source record, a reporting total, a policy finding and a later correction. A total of 61 against a capacity of 60 is not useful enough unless a reviewer can reproduce the count and determine whether the cause is an extra admission, a replayed upload, a withdrawal, a wrong programme mapping or a revised approval.

For a student, the central need is to see and challenge an institution's claim without being excluded by a missing identifier. For staff, it is to complete reporting without entering the same facts again. For a regulator, it is to distinguish incomplete evidence from a substantive concern. For a verifier, it is to obtain a narrow current assertion from an authorised source. These are **design hypotheses**, informed by public workflows; their frequency and value have not been measured through interviews.

The proposed product succeeds only if it reduces total reconciliation work while preserving students' correction rights. More dashboards or more alerts are not success measures by themselves.

2. Alternatives and the selection

Alternative	Useful strength	Main tradeoff	Decision
Replace the institutional ERP	One controlled workflow from application onward	High migration, training and integration burden; duplicates documented Samarth functions	Reject for this scope
Build an API-first national verification registry	Appealing single verification journey	Arbitrary-ID bulk lookup is not established; permission and source coverage remain external dependencies	Defer
Add SetuTrace beside existing systems	Imports existing records; preserves provenance, corrections and review	Mapping and reconciliation still cost time; operator governance is essential	Selected for evaluation
Improve spreadsheets, templates and checklists	Low setup cost; familiar to small institutions	Harder version control, access separation and correction propagation	Retain as baseline and possible final choice

The selected approach is feasible without government API access because its first value is local evidence reconciliation. It does not prove superior value. If a small institution can meet its needs with an improved checklist and less effort, that is the better outcome there. The product must earn its extra complexity in comparative testing. Gopika's opportunities; Kanishkar's feasibility base.

3. Product concept and defensible differentiation

SetuTrace centres on an **evidence card**, attached to each report finding or student record. It displays four separate questions: what the institution asserted; what an authorised provider returned; what the applicable rule evaluated; and what a responsible reviewer resolved. Format, source, revision and freshness are visible. A signature adds integrity and issuer attribution; it does not make an incorrect assertion true.

The proposed differentiator is a connected correction trail: source revision → rule evaluation → approved reporting snapshot → later correction → superseded output. Staff can explain a changed total; a student receives a case reference; an authorised reviewer can reconstruct an earlier decision. This is a product hypothesis, not a claim that competitors lack all such functions. The research did not exhaustively inspect every Samarth deployment or NAD integration.

Samarth remains the local operational system. NAD remains the appropriate authorised award infrastructure. SetuTrace's first receipt describes an institution's enrolment assertion and reporting state, not a degree, official AICTE acknowledgement or NAD publication. GOP-F01-F06; credential trust design.

4. Journeys designed around people

Institution staff and approver

An operator exports existing ERP data, chooses a saved mapping and previews changes before committing. Stable source keys distinguish a new enrolment from a replay or correction. Errors identify the field, reason and responsible action; staff do not repair the same value independently in two systems without a reconciliation record.

A separate approver reviews material differences, unresolved rows and the selected policy pack. Approval freezes an explicit snapshot. The dashboard distinguishes **locally accepted**, **export ready**, **submitted through an authorised route**, and **official acknowledgement evidenced**. A manually attached receipt records its uploader and provenance; it is not described as API readback. No current AICTE submission/readback contract has been established. Dharanish's import protocol; SUD operation matrix.

Student, including missing or disputed APAAR

A student enters through an institution-approved account or assisted identity process and sees their own institution claims, evidence status and correction history. Access must not depend circularly on possessing the very APAAR being disputed. A missing or foreign-student linkage route creates an internal reference and a pending case; record capture continues. Whether a final regulator submission may omit APAAR remains unresolved.

The student can challenge a wrong course, date or identifier and receives an owner, status and reasoned outcome. They cannot silently overwrite an institution or provider record. Assisted access uses limited, recorded delegation, accessible language and keyboard/screen-reader support; disability does not imply incapacity. No invented deadline or automatic admission denial fills an unresolved policy gap. Missing-ID research; permission design.

Regulator analyst and case reviewer

An analyst starts with authorised aggregate views: reported count, policy coverage, unresolved evidence, reporting cutoff and data freshness. Counts reveal their denominator and incomplete population. A potential capacity breach opens an accountable review, not an automatic cancellation of the last student added.

Cross-institution matching is conditional on documented regulator authority. Matching cases remain internal to assigned reviewers. Every institution receives a uniform batch receipt; neither a special ticket nor a changed notification reveals that another institution has a matching record. Clarification asks only about the institution's own facts. Timing, exports and support responses need the same protection. Residual inference and collusion remain risks. Security T03/T12.

Employer or other authorised verifier

A vetted verifier requests a purpose and limited fields. The student can approve or decline a separate disclosure grant naming recipient, fields, record revision and expiry. Declining does not alter admission. At retrieval, the service checks the recipient and current grant again and returns the source, assertion type, evidence date and status. A forwarded URL or QR alone grants nothing.

Withdrawal stops future service access, including pending exports once revocation commits; it cannot erase copies already downloaded. An unavailable status service produces an unavailable result, not a current-validity badge. Actual employer demand and institutional permission for this workflow remain unproven. NAD verifier principles, SUD-103; proposed disclosure lifecycle.

5. Rules, exceptions and correction lifecycle

Use a versioned policy pack scoped to institution, campus, programme, entry route, cohort/year, jurisdiction and effective period. Each rule cites its authority, required inputs and exception route. Missing current authority yields `POLICY_UNAVAILABLE`; a historical percentage is never silently activated. The inspected AICTE base handbook came through an institution mirror, and current amendment completeness remains unresolved. PRA-101-109; 14-rule catalogue.

Concurrent enrolments are not automatically invalid. UGC's modified guidance allows qualifying combinations and includes conditions for earlier programmes. Mode, timing, recognition and applicable council/university rules matter; the guidance excludes PhD. UGC 2025 modification, PRA-103, guidelines 1-5.

Nationality/status, candidature and seat entitlement remain separate. NRI, OCI, Foreign National, lateral-entry and other additional pools require their own applicable authority. There is no universal supernumerary percentage or shared extra-seat bucket. Worked cases explain why an apparent overlap or excess count can require correction rather than adverse action. POL-03-06/12 and C15-C20.

Correction follows **challenge → acknowledge → authenticate evidence → assign source owner → amend or explain refusal → re-evaluate → supersede affected outputs → appeal if contested**. Institution staff correct enrolment, the

provider corrects academic identity, and the awarding body corrects an award. SetuTrace tracks these actions; it cannot repair external systems by itself or promise their turnaround.

Store event-effective time separately from recording time. For the inspected Maharashtra scope, an authenticated online cancellation takes effect irrespective of later signed-copy receipt; an internal approver must not delay that legal effect. This is a scoped rule, not a national default. PRA-109, section 15(a), PDF p.13; C14.

6. Architecture and practical data flow

Use a modular application with a relational database and a separate worker, keeping the operational footprint understandable. The conceptual flow is: **institution export → quarantined staging → mapping/preview → revisioned institution evidence → rule evaluation → approver snapshot → reporting/export**. Student corrections re-enter the revision path. Conditional provider evidence and restricted regulator review attach to records through separate permission boundaries.

The core entities are institution, institution-local student, identity claim, restricted person link, programme offering, approval version, enrolment revision, import job, rule finding, correction case, submission snapshot and disclosure grant. Internal random identifiers preserve continuity when APAAR is missing or corrected. Names and unverified repeated IDs never trigger automatic person merges. Dharanish's entity model.

An upload token plus content/mapping fingerprint identifies a retry. Changed content under the same token conflicts; unchanged business records under a new token are no-ops. Corrections include an expected version, preventing a stale spreadsheet from restoring an old course or withdrawal state. Bounded chunks commit row outcomes, revisions, audit and outbox events atomically. Partial progress is visible and cannot masquerade as a completed snapshot. Duplicate-tolerant workers consume events and rebuild projections. Proposed import protocol; AWS outbox guidance, DHA-104.

The optional provider adapter binds one authorised account session to one local case. Requester v2.4 documents a token-account APAAR operation with no arbitrary-ID parameter and recommends one call after acquiring the token. Its response may include personal and academic fields; selective provider-side projection is not established. Assess the payload, derive minimal needed evidence and discard unrelated fields without logging the response. Deposited credits do not establish exhaustive current admissions. SUD-105, printed pp.16-19; inspected contract and ambiguities.

Provider calls occur outside database transactions. Denied permission, token failure, not-linked data, mismatch, outage and unexpected schemas remain distinct. Bound retries according to approved provider rules; expose staleness and preserve work. Never convert HTTP 404 or timeout into nonexistence or fraud. Institution acceptance does not depend on a successful provider call.

7. Complete proposed technology stack

These are engineering choices for a future implementation, not installed or tested project components. Pin supported versions and licences when implementation begins.

Area	Proposed choice	Purpose and limit
User interface	React, TypeScript, Vite, semantic HTML/CSS	Responsive staff/student views; accessible forms; no sensitive offline cache
API and schemas	Python, FastAPI, Pydantic; generated OpenAPI	Typed request validation and explicit evidence contracts
Persistence	PostgreSQL 18, SQLAlchemy, Alembic	Scoped constraints, transactions, revisions and reversible migration planning

Area	Proposed choice	Purpose and limit
Jobs and events	Separate Python worker; PostgreSQL jobs, leases and outbox	Durable bounded work; postpone Redis/Kafka until measured need
Imports and exports	Python CSV parser; read-only openpyxl XLSX later; JSON and controlled CSV/PDF outputs	CSV first; bounded parsing, versioned mappings and spreadsheet-formula protection
Authentication	Existing institutional OIDC provider; Keycloak for local synthetic setup; maintained Authlib client	Server session in secure HttpOnly cookies, CSRF protection, privileged MFA and role removal
Files and secrets	Private S3-compatible storage; managed KMS and secret manager for deployment	Scoped downloads; encryption; separate signing/encryption keys; no secrets in source
Development and CI	Git, locked dependencies, Docker Compose; GitHub Actions; Ruff, TypeScript checks, dependency and secret scans	Reproducible synthetic environment and reviewable changes
Testing	pytest, Hypothesis, real PostgreSQL/Testcontainers; Playwright and axe-core; Locust for future workload tests	Invariants, real transaction races, browser/accessibility and measured load
Deployment	Ubuntu LTS container host behind Caddy; managed PostgreSQL, private storage and backups for a permitted pilot	TLS, restricted network paths, migrations and restoration; resize before real use
Operations	OpenTelemetry; Prometheus/Grafana; redacted structured logs and separate protected audit events	Queue age, failures, projection lag and access denials; no student IDs in metric labels

FastAPI documents OpenAPI and validation capabilities; Vite supplies frontend build tooling. PostgreSQL RLS can add defence but owners and BYPASSRLS roles bypass ordinary policies: use separate migration roles and test non-owner application/worker roles. FastAPI, DHA-108; Vite, DHA-109; PostgreSQL 18 RLS, DHA-103.

Use a trusted issuer/key directory and maintained cryptographic libraries for an online signed assertion. Managed KMS can protect encryption/signing keys; policy, rotation and issuer mandate still require explicit design. No blockchain, LLM adjudicator or Kubernetes cluster is needed for the first loop. Keycloak OIDC; AWS KMS; credential lifecycle.

8. Security and legal boundaries

Authorisation is checked on every API, worker, file and export, using trusted institution membership, role, purpose and resource scope. Separate operator and approver roles; restrict support to masked metadata. Tenant controls must survive pooled connections, caches and queued jobs. Sensitive changes and exceptional access need accountable approval and audit. Permission matrix; OWASP authorisation, SUH-109.

Treat uploads as hostile input: limit bytes, rows and processing time; quarantine files; reject unsupported formats; never execute formulas/macros; protect exported spreadsheet text. Protect claimed identifiers separately. Any authorised matching token is keyed, access-controlled and still personal data; a plain hash is not anonymisation. Do not collect Aadhaar, biometrics, wallet PINs or OTPs in this proposed application. Threats T01-T09.

DPDP is phased. On 27 October 2026, its main processing, notice, consent, child-data and rights provisions remain in the future commencement stage; the project cannot claim blanket DPDP compliance. The legal review records the final Rules, December corrigendum and October 2026 order and preserves publication-date uncertainty. Current IT Act/SPDI applicability and the full SPDI Rules retrieval remain open review items. G.S.R.843(E), SUH-102; legal-status assessment, SUH-101-116.

A pilot needs a named operator, purpose/field/recipient authority map, processor agreements, retention schedule, age/guardian process, grievance route and incident responsibility. Institution reporting authority, provider permission and employer disclosure are different decisions. Applicable CERT-In incident/log obligations must be assessed separately; an educational purpose is not blanket sharing permission. SUH-106 and pilot decisions.

9. MVP boundaries and requirement coverage

The first future MVP is **one synthetic end-to-end correction and reporting loop** across two fictional tenants and one scoped rule pack. Include import, mapping, preview, approver snapshot, student challenge, rule explanations, export, audit and recipient-bound demonstration verification. Display synthetic provider/issuer labels throughout. This report does not implement that MVP.

Requirement	First demonstrator	Conditional next step	Acceptance evidence
R01 Upload/update	Scoped imports and revisions	Approved institutional contract	Retry and stale-edit tests
R02 Required dimensions	Separate intake, category, route and pool fields	Current course/year approval	Reviewed field map and denominator
R03 Detection	Deterministic data/policy findings	Authorised restricted overlap review	Boundary and exception cases
R04 Dashboards	Own-tenant and fictional regulator views	Approved real reporting scope	Reconciled totals and freshness
R05 Verification	Recipient-bound synthetic assertion	Vetted recipient and authority	Denial/revocation checks
R06 Verifiable records	Revision, provenance, demo signature/status	Trusted issuer mandate	Tamper, stale-status and key tests
R07 Integration	CSV/internal contracts; provider fixture	Approved provider/AICTE adapters	Contract and authorised test evidence
R08 Protection	Roles, grants, audit and threat fixtures	Legal/security entry gates	Independent scoped assessment
R09 Reports	Versioned totals and evidence export	Accepted regulator/accreditation definitions	Reproduction from snapshot
R10 Certificate IDs	Synthetic awarding-body template preview	Confirmed issuer/display requirements	Identifier distinction/privacy review

R10 does not justify public identifier retrieval through a QR. Awarding bodies control real certificate printing; APAAR, institutional student ID and AICTE enrolment number remain distinct. Direct Aadhaar integration, universal ERP connectors, nationwide automatic enforcement, predictive fraud scores and W3C credential interoperability are outside the first MVP. Later work requires demonstrated need and authority. POL-14/C22; SUD access ladder.

10. What the evidence actually establishes

The assembled desk research establishes useful constraints: existing import/award systems, a documented account-linked provider operation, scoped policy exceptions, security design requirements and unresolved access dependencies. It does not establish adoption, saved time, accuracy on real students or operational compliance.

The independently reproduced original model contains **1,000 fictional rows**, a separate fictional reference cohort and twelve sequential model checks. Eight comparisons with preserved original outputs matched. Rules identified 80 labelled cases with zero false alerts in that constructed dataset; adding the statistical signal found ten additional labelled cases and added ten false alerts. Its incremental precision was therefore 50%. This supports a rules-first choice, not a real-world accuracy claim. Reproduction README; recorded results.

Replay/correction behaviour in that model is not a database, concurrency or recovery test of the proposed product. The coarse four-millisecond observation excludes parsing, storage, network and the user workflow. No throughput, national scalability, secure tenancy or fraud-detection claim follows. Optional anomaly triage stays deferred until representative, held-out evaluation justifies the extra review burden.

Supported key points and proposed solutions

Problem addressed	Proposed solution	What is validated now
Replayed or corrected exports alter totals	Idempotent import, scoped keys and revision trail	Sequential fictional model reproduced; real database behavior untested
A valid-looking ID is treated as admission proof	Separate format, provider, institution and policy evidence	Provider contract and ecosystem boundaries inspected; live linkage untested
Legitimate concurrent study is wrongly flagged	Contextual rule and human correction route	Current UGC guidance inspected; pilot applicability still needs review
Historical intake assumptions cause false findings	Versioned approval and rule pack; UNKNOWN when incomplete	Base policy and a real corrigendum inspected; complete amendments unresolved
Corrections fail to reach reports or verifiers	Superseded snapshots and revision-bound online assertions	Design and threat analysis only; no product propagation test
Matching or QR lookup leaks student information	Uniform receipts, authorised internal review and scoped grants	Threat mechanism and controls reasoned; implementation security untested
Staff and students repeat reconciliation work	Reuse exports and give each correction an owner and trail	Existing reference workflows documented; time savings and adoption unmeasured

The first three rows have supporting source or model evidence, not a field-tested solution. The requirement matrix records acceptance gates for every feature.

11. Workload, operating effort and cost

For planning only, target a later demonstrator at 10,000 records, 20 concurrent users and uploads capped at 10 MB or 10,000 rows, whichever comes first. Only the 1,000-row model has been executed. A sizing sketch of one million records × 2 KB × three revisions gives 6 GB; adding 1 GB ancillary payload and assuming 2× overhead gives 14 GB, before current rows, WAL, backups or replicas. This is arithmetic, not a storage measurement. Dharanish sizing assumptions.

Research priced an illustrative Linux 2 GB public-IPv4 Lightsail VM at USD12/month and an encrypted 2 GB standard database at USD30/month: USD42/month for those two components. The smaller USD15 database lacks encryption. These are list-price examples, not a pilot quotation or a capacity recommendation. AWS Lightsail pricing, DHA-106.

Add object storage, keys, logs, backups, network, identity service, notifications, support, taxes and staff effort. The complete proposed stack needs fresh sizing and region/contract review. Provider pricing/access is unknown. No resource has been purchased. The largest uncertain operating cost may be reconciliation: for an assumed 1% of 10,000 records requiring ten minutes each, review takes about 16.7 staff-hours before approvals or follow-up. Measure this input rather than hiding it behind cheap hosting.

12. Staged validation and stop conditions

Stage A - evidence and workflow confirmation. Obtain official SIH wording, a willing accountable institution, blank templates, current approval/amendment chain and permission to study tasks. Observe a small, diverse sample of operators, approvers and students, including missing-ID and assisted-access cases. Map actual corrections and compare the improved-checklist baseline. Interviews are qualitative, not national prevalence estimates. Stop product expansion if the workflow merely duplicates entry or lacks a responsible correction owner.

Stage B - synthetic implementation, if authorised later. Instantiate all 22 policy cases and provider failure fixtures; execute real database races, interrupted jobs, malformed uploads, tenant denials, revocation/export races and restore tests. Entry to real data requires zero known critical access failures, no silent lost updates, reproducible snapshot totals and visible unresolved states. Demonstrate that comparable matching/nonmatching submissions reveal no match bit across response, notification, export and timing channels; investigate residual inference. These are proposed criteria, not passed checks.

Stage C - permitted shadow pilot. Start with one institution/programme/cohort, two separated staff roles and a student grievance contact. Keep official workflow and rollback available. Add a second institution only after matching authority and access boundaries are approved. Live provider linkage requires sponsor/use-case approval, exact scopes, authorised test arrangements and acceptance: the partner SOP excludes temporary testing access. SUD-106, section 5(vi).

Measure total operator, approver and student effort; correction completion; explanation comprehension; false-alert burden; exclusion complaints; reconciliation accuracy; and willingness to continue. Proposed progression targets are at least 90% successful completion of essential tasks without facilitator correction, at least 20% lower median total staff time than the baseline, and no increase in unresolved student cases. Agree targets beforehand; report sample size, distributions and failures. A small pilot cannot certify national effectiveness.

Stage D - controlled expansion. Recheck policy/legal commencement, complete independent security assessment, test restore and key-compromise procedures, and obtain genuine institutional acceptance. Expand one integration or cohort at a time. Stop affected processing on unauthorised disclosure, wrong-person linkage, failed revocation, silent overwrite, unreviewable policy, or irreversible wrongful rejection. Contain the event, preserve restricted evidence, use the manual route and reopen only after documented human review.

SetuTrace is defensible when it helps people correct and explain records without overstating what identity evidence proves. The current deliverable is the research-backed design and its testable assumptions. Actual AI critique and resolutions remain separate from the human decisions still required.

Lead publication update: The final remote refresh incorporated newer Pranav and Sudhir contributions, preserved alongside their specialist reconciliations in the assignment audit.